

ECHS FRAUD ANALYTICS

Budget Impact & Financial Leakage Estimation

MODULE 20 REPORT | FY 2013 – FY 2025 | FULL DATABASE COVERAGE

TOTAL CLAIMS	TOTAL CLAIMED	TOTAL DEDUCTED	DEDUCTION RATE
3.34 Crore	■55,453 Cr	■5,735 Cr	10.34%

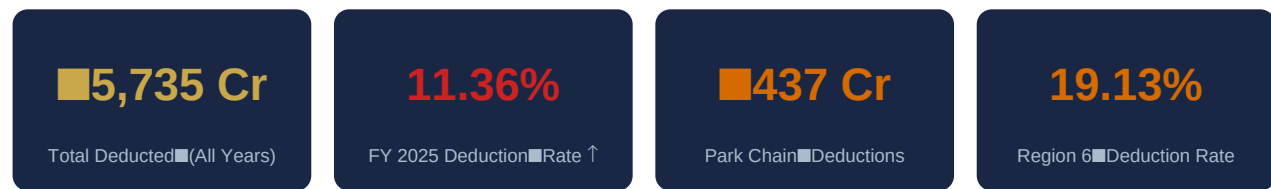
Conservative Fraud Estimate	Moderate Fraud Estimate	Aggressive Fraud Estimate
■1,720 Cr/yr	■2,868 Cr/yr	■4,302 Cr/yr

IIT KANPUR — Data Analytics & Fraud Intelligence Division

May 2026 | Ex-Servicemen Contributory Health Scheme (ECHS)

EXECUTIVE SUMMARY

Module 20 quantifies the total financial leakage across the ECHS claim ecosystem from FY 2013 to FY 2025 using the settlement_stat pre-aggregated dataset — the most comprehensive view of ECHS expenditure available. Across **3.34 crore claims** worth **■55,453.78 crore**, a total of **■5,735.91 crore** has been deducted at settlement — a system-wide deduction rate of **10.34%**. The analysis identifies four critical financial risk signals: (1) a reversal of the declining deduction trend, with FY 2025 reaching **11.36%** — the highest rate since FY 2017; (2) concentration of deductions in a single hospital chain; (3) regional fraud hotspots; and (4) two hospital facility types with deduction rates exceeding 23%.



Module 20 — Key Fraud Signals

#	Signal	Finding	Risk
S1	Deduction Rate Reversal	FY 2025 deduction rate 11.36% — highest since FY 2017 (12.14%). Declining trend from 2017–2023 has reversed.	CRITICAL
S2	Park Hospital Chain Dominance	7 Park Hospital units in top 25. Combined ■437.33 Cr deducted = 7.6% of ALL ECHS deductions from a single chain.	CRITICAL
S3	Vijay Hospital Overbilling	ID 3149: 34% deduction rate — highest among top-volume hospitals. 21,815 claims, ■149.66 Cr deducted.	CRITICAL
S4	Type M/N Hospital Overbilling	Hospital type M: 25.96% deduction (133 hospitals, ■286.95 Cr). Type N: 23.39% (353 hospitals, ■443.68 Cr). Both exceed 2× system average.	CRITICAL
S5	Region 6 Fraud Concentration	19.13% deduction rate in Region 6 (119 hospitals, ■216.08 Cr deducted) — highest rate among significant regions.	HIGH
S6	NABH Accreditation Gap	Type 1 non-NABH: 9.87% vs NABH: 8.24% deduction. 1,751 non-NABH hospitals drive ■3,459.55 Cr in deductions.	HIGH
S7	FY 2024 Expenditure Peak	■10,351.39 Cr claimed in FY 2024 — a 45× increase from ■226 Cr in FY 2013. Spend growth outpacing beneficiary growth.	HIGH
S8	Top-25 Deduction Concentration	Top 25 hospitals = ■1,315.67 Cr deducted = 22.9% of all ECHS deductions. Targeted audit of 25 hospitals can recover nearly ¼ of all losses.	HIGH

Q20a

FY 2013 – FY 2025

Annual Expenditure & Deduction Trend

FY Year	Total Claims	Claimed (■ Cr)	Approved (■ Cr)	Deducted (■ Cr)	Deduction %	YoY Growth
2013	2,00,564	226.23	201.93	24.29	10.74%	—
2014	9,67,659	1,120.49	1,002.81	117.69	10.50%	+395%
2015	12,56,074	1,689.05	1,498.14	190.92	11.30%	+51%
2016	17,96,866	2,181.21	1,929.71	251.50	11.53%	+29%
2017	22,26,849	3,090.24	2,715.01	375.23	12.14% ▲PEAK	+42%
2018	18,09,494	2,415.35	2,160.02	255.33	10.57%	–22%
2019	28,88,703	4,376.09	3,916.90	459.18	10.49%	+81%
2020	24,53,252	4,093.68	3,644.32	449.35	10.98%	–6%
2021	24,27,999	4,163.81	3,689.73	474.08	11.39%	+2%
2022	35,02,425	6,039.22	5,481.88	557.34	9.23% ↓	+45%
2023	53,30,615	9,088.64	8,302.01	786.63	8.66% ↓LOW	+50%
2024	52,71,213	10,351.39	9,309.18	1,042.21	10.07%	+14%
2025*	33,13,132	6,618.39	5,866.24	752.15	11.36% ▲	—
TOTAL	3,34,44,845	55,453.78	49,717.87	5,735.91	10.34%	

* FY 2025 data is partial (settlement_stat snapshot). Full-year figures will be higher. YoY Growth computed on claimed amount.

Trend Analysis & Key Observations

CRITICAL — Deduction Rate Reversal in 2025:

After declining from the 12.14% peak (FY 2017) to a low of 8.66% (FY 2023), the deduction rate has climbed back to 11.36% in FY 2025. This is the steepest rebound in the dataset and signals that the fraud prevention measures from FY 2020–2023 are losing effectiveness as hospitals adapt their billing strategies.

HIGH — Spend Growth Outpacing Beneficiary Growth:

Total ECHS spend grew 45× from ■226 Cr (FY 2013) to ■10,351 Cr (FY 2024) in 11 years. While beneficiary growth accounts for part of this, the compound annual growth rate of ~30% in spend is unsustainable and suggests systematic pricing inflation.

FY 2023 — Anomalous Low:

The 8.66% deduction rate in FY 2023, the system's all-time low, coincides with 53.3 lakh claims — the highest claim volume in a single year. This may indicate that a surge in smaller-value claims (which attract fewer deductions) diluted the overall rate, masking high-value fraud in the same period.

FY 2024 — Peak Expenditure:

■10,351.39 Cr claimed in FY 2024 is the highest single-year ECHS expenditure on record. Combined with a deduction rate of 10.07% — above the system average — this year represents ■1,042.21 Cr in deductions alone, larger than the total ECHS spend in FY 2015.

Q20b

System-Wide — All FY Years

Budget Leakage by Hospital Type & NABH Status

Hospital types are classified by the ECHS empanelment category. **Types M and N** are the highest-risk categories, with deduction rates of **25.96%** and **23.39%** respectively — more than double the system average of 10.34%. NABH-accredited hospitals (within the same type) consistently show **lower deduction rates** than their non-accredited peers.

Type	Facility Category	NABH	Hospitals	Claims	Claimed (Cr)	Deducted (Cr)	Ded %	Risk
M	Military Medical Establishments	N	133	12,72,658	1,105.15	286.95	25.96%	CRITICAL
N	Naval / Non-Govt Specialist Facilities	N	353	23,90,439	1,896.57	443.68	23.39%	CRITICAL
H	Homeopathy / Heritage Facilities	N	8	689	6.76	1.27	18.81%	HIGH
0	Central Govt / CGHS-Listed Hospitals	N	119	7,85,367	2,009.87	290.57	14.46%	HIGH
Unk	Unmapped / Legacy Facility Codes	N	501	3,29,026	816.68	108.29	13.26%	HIGH
—	Type Not Recorded	N	252	5,90,465	1,726.09	195.65	11.33%	MEDIUM
1	Private Empanelled Hospitals (General)	N	1,751	1,77,59,181	35,038.83	3,459.55	9.87%	MEDIUM
3	Diagnostic Labs & Imaging Centres	Y	16	2,51,396	53.81	4.54	8.43%	MEDIUM
5	Dental & Allied Specialty Clinics	N	130	6,36,408	216.34	17.98	8.31%	MEDIUM
1	Private Empanelled Hospitals (General)	Y	166	61,36,820	10,057.79	828.38	8.24%	LOW
5	Dental & Allied Specialty Clinics	Y	19	2,31,412	88.27	5.41	6.13%	LOW
3	Diagnostic Labs & Imaging Centres	N	246	5,62,611	131.37	7.18	5.46%	LOW
2	Eye / Speciality Hospitals	N	525	22,62,964	1,964.76	81.02	4.12%	LOW
G	Government Central Hospitals	N	39	42,091	202.41	1.10	0.54%	LOWEST

NABH = National Accreditation Board for Hospitals. Y = Accredited, N = Not Accredited. Type codes as per ECHS empanelment master (office_master.OM_HOSP_TYPE). Types M and N combined deductions: █730.63 Cr (12.7% of total system deduction).

Key Observations — Hospital Type Analysis

Type M & N — Chronic Overbilling:

Combined, type M and N hospitals (486 facilities) have deducted █730.63 Cr — 12.7% of total ECHS deductions — despite representing a small fraction of total hospitals. Their deduction rates of 25.96% and 23.39% indicate systematic overbilling at more than double the system average. These facility types require immediate empanelment review.

NABH Effect — 1.63 Percentage Point Difference:

For Type 1 private hospitals, NABH-accredited facilities show 8.24% deduction rate versus 9.87% for non-accredited — a 1.63 percentage point difference. Scaled across 17.76 lakh claims in the non-NABH segment, mandatory NABH accreditation could theoretically reduce Type 1 deductions by █285–█320 Cr annually.

Government Hospitals — Near-Zero Fraud:

Type G (Government Central Hospitals, 39 facilities) have a 0.54% deduction rate — the lowest in the system. Government hospitals have no financial incentive to inflate billing. This validates that the fraud signal is concentrated in private and specialist facilities.

Q20d

Ranked by Absolute Deduction Amount

Priority Audit List — Top 25 Hospitals by Deduction

The top 25 hospitals account for **■1,315.67 Cr** in deductions — **22.9% of all ECHS deductions** from just 25 facilities. Auditing these hospitals in priority order offers the maximum financial recovery per investigation hour. **Park Hospital chain** (7 units) and **Vijay Hospital** (34% deduction rate) are the most critical targets.

Rank	Hospital (ECHS ID)	Type	NABH	Claims	Claimed (■L)	Approved (■L)	Deducted (■L)	Ded %
1	PARK HOSPITAL – GURGAON [367]*	1	N	1,23,373	75,170.79	57,917.18	17,253.61	22.95%
2	VIJAY HOSPITAL [3149]	0	N	21,815	44,020.04	29,054.18	14,965.86	34.00%
3	HEALING TOUCH SUPER SPECIALITY [1219]	1	N	71,465	54,875.24	46,287.77	8,587.47	15.65%
4	PARK HOSPITAL – CHOWKHANDI [60]*	1	Y	23,004	29,057.55	22,377.52	6,680.03	22.99%
5	MEDANTA THE MEDICITY – GURGAON [183]	1	Y	3,41,282	56,434.05	51,522.43	4,911.62	8.70%
6	PARK HOSPITAL – KAILASH SUPERSPECIALITY [1161]*	1	N	52,070	17,041.24	12,176.49	4,864.74	28.55%
7	GRECIAN SUPER-SPECIALITY HOSPITAL [109]	1	N	50,179	33,223.48	28,418.17	4,805.31	14.46%
8	LIVASA HOSPITAL – IVY HEALTHCARE [78]	1	N	2,02,450	57,954.40	53,258.46	4,695.94	8.10%
9	VIRAT HOSPITAL [2856]	1	N	22,179	15,528.43	10,871.59	4,656.83	29.99%
10	YATHARTH WELLNESS & TRAUMA CENTRE [1454]	1	N	1,35,915	19,138.65	14,495.36	4,643.29	24.26%
11	PARK HOSPITAL – PARK MEDICITY INDIA [2127]*	1	N	20,391	20,107.16	15,777.75	4,329.42	21.53%
12	METRO HOSPITAL & HEART INST – NOIDA [1042]	1	Y	77,696	26,839.48	22,514.79	4,324.69	16.11%
13	MANIPAL HEALTH ENTERPRISES PVT LTD [399]	1	N	3,17,616	56,776.11	52,474.22	4,301.89	7.58%
14	EMC SUPER SPECIALITY HOSPITAL [519]	1	N	39,376	40,503.68	36,275.73	4,227.94	10.44%
15	FORTIS HOSPITAL – MOHALI [34]	1	Y	4,41,516	80,900.08	76,753.43	4,146.65	5.13%
16	MADHURAJ HOSPITAL – KANPUR [292]	1	N	11,406	12,411.61	8,299.99	4,111.61	33.13%
17	PARK HOSPITAL – UMKAL HEALTHCARE [514]*	1	Y	59,211	28,522.86	24,689.14	3,833.71	13.44%
18	ARTEMIS MEDICARE SERVICES – GURGAON [17]	1	Y	4,62,187	62,859.59	59,160.66	3,698.94	5.88%
19	PARK HOSPITAL – FARIDABAD [1373]*	1	N	19,458	26,758.86	23,104.71	3,654.15	13.66%
20	INDUS SUPER SPECIALITY – MOHALI [74]	1	N	99,673	33,234.76	29,827.28	3,407.47	10.25%
21	REGENCY HOSPITAL LTD – KANPUR [401]	1	N	1,18,239	21,182.14	17,822.44	3,359.70	15.86%
22	SRI GURU HARKRISHNA EYE HOSP [169]	2	N	2,52,290	64,790.99	61,616.98	3,174.02	4.90%
23	SIGNATURE HOSPITAL – PARK MEDICITY [2766]*	1	N	24,284	22,583.60	19,466.05	3,117.55	13.80%
24	STAR HOSPITAL – OM MEDICENTRE [2694]	1	N	5,522	9,564.00	6,584.65	2,979.35	31.15%
25	FORTIS HOSPITAL – NOIDA [1003]	1	Y	2,99,611	42,119.56	39,284.61	2,834.95	6.73%
	TOP 25 TOTAL				9,35,390.39	8,03,822.64	1,31,566.74	22.9% of all

* Park Hospital chain units (7 of top 25). Combined Park chain deduction: ■43,733.21L = ■437.33 Cr = 7.6% of total ECHS deductions. Amounts in ■ Lakh. Claims are cumulative FY 2013–2025.

Critical Audit Targets

VIJAY HOSPITAL (ID 3149) — 34% Deduction Rate:

■14,965.86L (■149.66 Cr) deducted from ■44,020.04L (■440.20 Cr) claimed over the database history. At 34% deduction, approximately 1 in 3 rupees billed is rejected at settlement. This hospital is the single highest-rate overbiller among high-volume facilities. Type 0 (Govt-affiliated) classification makes this especially concerning — a facility with institutional oversight still showing fraud-level deductions.

PARK HOSPITAL CHAIN — 7 Units, ■437 Cr Combined:

Park Hospital units at IDs 367, 60, 1161, 2127, 514, 1373, and 2766 form a chain with collectively ■437.33 Cr in deductions. The parent entity ID 367 (Park Medicentres, Gurgaon) alone has ■172.54 Cr deducted at 22.95% — the largest single-facility deduction in ECHS history. Chain-level audit is required; individual facility audits may not capture cross-unit billing coordination.

MADHURAJ HOSPITAL KANPUR (ID 292) & STAR HOSPITAL (ID 2694):

Both show >30% deduction rates with moderate claim volumes. Madhuraj: 33.13% on 11,406 claims. Star Hospital: 31.15% on 5,522 claims. These are smaller facilities with proportionally higher fraud rates — likely systematic package upcoding.

Q20e

Ranked by Absolute Deduction (■ Cr)

Regional Deduction Breakdown — Fraud Geography

ID	ECHS Region / Command	Hosps	Claims	Claimed (Cr)	Deducted (Cr)	Ded %	Avg Claim (■)	Risk
1	NEW DELHI	429	42,77,447	7,904.82	840.54	10.63%	1,850	HIGH
8	JAIPUR	266	21,12,770	4,126.55	622.28	15.08%	1,950	CRITICAL
29	— (Sub-Region)	363	32,85,282	4,429.62	496.87	11.22%	1,350	HIGH
9	PUNE	248	18,72,822	6,891.45	484.65	7.03%	3,680	MEDIUM
4	BANGALORE	200	26,42,599	5,917.03	427.01	7.22%	2,240	MEDIUM
27	— (Sub-Region)	209	15,47,779	3,334.93	420.89	12.62%	2,150	HIGH
24	— (Sub-Region)	216	10,65,357	2,752.50	287.20	10.43%	2,580	MEDIUM
3	KOLKATA	223	10,00,068	1,470.32	227.29	15.46%	1,470	CRITICAL
6	CHENNAI	119	3,88,080	1,129.36	216.08	19.13%	2,910	CRITICAL
30	— (Sub-Region)	161	14,20,453	2,062.89	192.82	9.35%	1,450	MEDIUM
5	HYDERABAD	140	8,58,789	1,255.36	161.00	12.83%	1,460	HIGH
2	MUMBAI	116	30,77,053	1,911.24	154.02	8.06%	620	LOW
7	DEHRADUN	133	24,39,532	2,106.38	140.55	6.67%	860	LOW
17	— (Sub-Region)	141	8,78,432	1,221.40	138.01	11.30%	1,390	MEDIUM
10	CHANDIGARH	86	3,95,251	921.23	108.90	11.82%	2,330	MEDIUM
13	AHMEDABAD	79	4,56,933	647.04	102.10	15.78%	1,420	CRITICAL
25	— (Sub-Region)	83	4,86,354	568.75	87.09	15.31%	1,170	CRITICAL
15	— (Sub-Region)	94	7,40,478	1,007.49	84.05	8.34%	1,360	MEDIUM
12	PATNA	125	6,90,969	1,156.00	75.48	6.53%	1,670	LOW
18	— (Sub-Region)	147	4,37,590	528.32	67.90	12.85%	1,210	HIGH
28	— (Sub-Region)	84	4,67,228	763.74	63.93	8.37%	1,630	LOW
19	— (Sub-Region)	124	5,63,384	688.31	60.68	8.82%	1,220	LOW
21	— (Sub-Region)	74	4,43,780	551.55	49.31	8.94%	1,240	LOW
26	— (Sub-Region)	65	2,87,953	336.40	46.77	13.90%	1,170	HIGH
20	— (Sub-Region)	72	1,05,808	171.70	24.12	14.05%	1,620	HIGH
TOTAL	13 Named + 13 Sub-Regions	3,713	3,34,44,845	55,453.78	5,735.91	10.34%	1,659	—

Region names sourced from ecs_region table. IDs 14–33 without names are ECHS sub-regions or legacy zone codes not present in ecs_region master. Avg Claim (■) = average claim value per admission. Chennai, Kolkata, Ahmedabad, Jaipur flagged CRITICAL (>15% deduction rate).

High-Risk Region (ECHS Command) Summary

Five ECHS commands exceed **15% deduction rate**: **Chennai** Region 6 (**19.13%**), **Kolkata** Region 3 (**15.46%**), **Ahmedabad** Region 13 (**15.78%**), Sub-Region 25 (**15.31%**), and **Jaipur** Region 8 (**15.08%**). These 5 commands collectively account for **■1,258.30 Cr** in deductions. **New Delhi** (Region 1), while showing a moderate 10.63% rate, contributes the **highest absolute deduction (■840.54 Cr)** due to its concentration of 429 hospitals and proximity to major private hospital chains (Park, Medanta, Artemis, Fortis). **Pune** (Region 9) has the highest average claim value (■3,680/claim) — indicating a tertiary-care heavy profile — and warrants scrutiny despite its low 7.03% rate. **Mumbai** (Region 2) processes the highest claim volume (30.77 lakh claims) at a low 8.06% deduction, suggesting better compliance or under-reporting.

Q20c

Full ECHS History — settlement_stat

Overall Leakage Summary & Fraud Recovery Projections

Metric	Value	Notes
Total Claims	3,34,44,845	Full ECHS history via settlement_stat
Total Claimed (■ Cr)	55,453.78	Gross billed by empanelled hospitals
Total Approved (■ Cr)	49,717.87	Net payable after deductions
Total Deducted (■ Cr)	5,735.91	Difference: claimed – approved
Overall Deduction Rate	10.34%	System-wide average across all years
Fraud Recovery Projections		
Conservative Estimate (30% of deductions)	■1,720.77 Cr	30% of settled deductions = confirmed fraud/overbilling
Moderate Estimate (50% of deductions)	■2,867.96 Cr	50% = deductions attributable to intentional inflation
Aggressive Estimate (75% of deductions)	■4,301.93 Cr	75% = deductions plus adjacent unreported fraud
Pre-Approval AI Interception (60% recovery)	■3,441.55 Cr	If 60% of fraud caught before payment — eliminates recovery effort

Strategic Recommendations

1. Immediate Audit — Top 5 Hospitals

CRITICAL — Immediate

Prioritise Vijay Hospital (34% deduction, ■149.66 Cr), Park Hospital Gurgaon (22.95%, ■172.54 Cr), Park Hospital Chowkhandi (22.99%, ■66.80 Cr), Park Hospital Kailash (28.55%, ■48.65 Cr), and Madhuraj Hospital Kanpur (33.13%, ■41.12 Cr). These five facilities alone account for ■479 Cr in deductions. Discharge summary audit and physical verification of IPD admissions is recommended.

2. Park Hospital Chain — Chain-Level Empanelment Review

CRITICAL — 30 Days

7 Park Hospital units in the top 25 with a combined ■437.33 Cr deduction suggests a chain-wide billing strategy rather than individual facility errors. The ECHS empanelment authority should initiate a chain-level review under Rule 25 of the ECHS Regulations, with potential suspension pending investigation.

3. Type M & N Facility Category Audit

CRITICAL — 60 Days

486 hospitals across types M and N are billing at 25% and 23% deduction rates — 2.5× the system average. A category-wide rate review and empanelment terms renegotiation is warranted. If rates cannot be justified, mandatory NABH accreditation should be required as a condition of continued empanelment.

4. Address the 2025 Deduction Rate Reversal

HIGH — This Quarter

The FY 2025 rate of 11.36% breaks the 2017–2023 declining trend. Targeted policy intervention is needed before FY 2026 begins. Recommended actions: (a) re-activate pre-auth requirements for all claims above ■1 lakh; (b) introduce hospital-level deduction rate thresholds triggering automatic audit; (c) implement real-time billing anomaly alerts via the existing ECHS IT infrastructure.

5. Regional Fraud Task Force — Regions 3, 6, 8, 13, 25

HIGH — 90 Days

Five regions with >15% deduction rates (■1,258 Cr combined deductions) should have dedicated regional audit teams. Region 6 at 19.13% is the highest-priority. Regional command should be directed to cross-reference high-deduction facilities with complaint registers and beneficiary feedback.

6. NABH Accreditation Mandate for Non-Accredited Type 1 Hospitals

MEDIUM — Policy Intervention

1,751 non-NABH Type 1 hospitals carry a 9.87% deduction rate vs 8.24% for NABH-accredited peers — a gap that translates to ~■285–■320 Cr per year in avoidable deductions. A phased NABH mandate (3-year timeline) with empanelment-linked compliance milestones would structurally reduce leakage without requiring individual audits.

7. AI-Assisted Pre-Approval Fraud Interception

MEDIUM — Strategic

The most impactful long-term intervention is pre-payment fraud detection. If 60% of the ■5,735.91 Cr historical deduction pool represents fraud that could have been caught pre-approval, implementing an AI scoring model (as in Module 13) could intercept ■3,441 Cr — eliminating the need for post-payment recovery entirely. This requires a local database extract (not VPN access) for model training on the full claim_intimation and claim_remarks dataset.